
UCL COMP0078: Supervised Learning

Problem Sheet

Technique Review

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If you find any mistakes/typos, please contact sl-support@cs.ucl.ac.uk.

Overview of Problems

Disclaimer: The problems I cover here are more difficult. I have included this selection of problems as they often use a creative use of what is covered in lecture and they are interesting some of the questions I put in here are inspired by question in class or what students asked me after class.

- **Problem 1:** Review on how to use Lipschitz conditions
- **Problem 2:** Deriving the mistake bound for Support Vector Machines (SVMs).
- **Problem 3:** Investigating the effects of averaging in gradient descent iterations.
- **Problem 4:** Analyzing variable step-size updates for perceptions.

Problem 1

Let $X_1, \dots, X_n$ be a sample from a distribution P with density p over the interval $[0, 1]$. The goal of nonparametric density estimation is to estimate p with as few assumptions about p as possible. We denote the estimator by $\hat{p}$. The estimator will depend on a smoothing parameter h , and choosing h carefully is crucial. To emphasize the dependence on h , we sometimes write $\hat{p}_h$. A simple and intuitive way to estimate $\hat{p}_h$ is by using histograms. To construct a histogram estimator, the range of the data is divided into m non-overlapping bins (or intervals) of equal width h . Let B_j represent the j -th bin, where:

$$B_j = \{x : a + (j-1)h \leq x < a + jh\}, \quad j = 1, \dots, m,$$

and $h = \frac{\text{range of data}}{m}$ is the bin width. The histogram density estimator is defined as:

$$\hat{p}_h(x) = \sum_{i=1}^n \frac{\hat{\theta}_i}{h} 1_{B_j}(x),$$

where $1_{B_j}(x)$ is an indicator function that equals 1 if $x \in B_j$, and 0 otherwise and $\hat{\theta}_i = \frac{1}{n} \sum_{i=1}^n 1_{B_j}(X_i)$.

1. A common loss function to use for nonparametric density estimation is the L_1 norm

$$\|\hat{p}(x) - p(x)\|_1 = \int |\hat{p}(x) - p(x)| dx$$

The reason why is because

- (i) L_1 norm is equivalent to the total variation loss. That is

$$\|\hat{p}(x) - p(x)\|_1 = 2 \sup_A |\hat{P}(A) - P(A)| = 2d_{TV}(\hat{p}, p)$$

where P is the probability measure induced by the density p .

(ii) and L_1 norm is unaffected by one-to-one smooth function. That is

$$\int_X |\hat{p}(x) - p(x)| dx = \int_{T^{-1}(Y)} |\tilde{\hat{p}}(y) - \tilde{p}(y)| dy$$

Prove these 2 statements.

2. Suppose that the probability density for distribution P is L -lipschitz show that

$$|\hat{p}(x) - p(x)| \leq Lh$$

and obtain bound on the L_1 risk.

3. Suppose our input space is now a unit cube $\mathcal{X} = [0, 1]^d$, provide an upper bound on the L_1 loss for this case using a similar analysis as previously done. (Hint: d will need to appear in the upper bound)

Problem 2

Suppose we have a set $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ of labeled examples in $\mathbb{R}^d$. It is NP-hard to find a linear separator that minimizes the number of points misclassified, so learning algorithms optimize other related quantities that are easier to solve. Support Vector Machines (SVM) solves the optimization problem:

$$\min_{\theta} \|\theta\|^2 + C \sum_{t=1}^n \zeta_t$$

subject to:

$$\begin{aligned} y_t(\theta^\top \mathbf{x}_t) &\geq 1 - \zeta_t \quad \forall i, \\ \zeta_t &\geq 0 \quad \forall i. \end{aligned} \tag{1}$$

Suppose that $\mathcal{D}$ has the property that the total distance one would need to move the points in order to make them separable by margin γ is d_γ . Show that when $C = \frac{2}{\gamma}$, the number of misclassification (i.e. mistake bound) made on $\mathcal{D}$ by the separator produced by the SVM is at most:

$$\frac{1}{2} + \frac{d_\gamma}{\gamma}.$$

Hint: We can find the value of d_γ with the following optimization problem:

$$\min_{\theta, \tilde{\gamma}_t} \sum_{t=1}^n \tilde{\gamma}_t$$

subject to:

$$\begin{aligned} \|\theta\| &= 1, \\ y_t(\theta^\top \mathbf{x}_t) &\geq \gamma - \tilde{\gamma}_t \quad \forall i. \end{aligned} \tag{2}$$

Here, $\tilde{\gamma}_t$ indicates the distance between point $\mathbf{x}_t$ and the appropriate margin. Also, note that a misclassification in SVM requires $\zeta_t \geq 1$, meaning each such mistake adds at least C to the SVM objective function.

Problem 3

In Week 4, you were introduced to subgradient descent, where it was demonstrated that for a function f that is convex, ρ -Lipschitz, and for which there exists a minimizer $\mathbf{w}^* \in \arg \min_{\mathbf{w}: \|\mathbf{w}\| \leq B} f(\mathbf{w})$ for some $B \in \mathbb{R}$, the algorithm has a convergence rate of $O\left(\frac{1}{\sqrt{T}}\right)$, where T is the number of iterations. Now, consider a modification: instead of focusing solely on the last iterate $\mathbf{w}_T$, we analyze the performance of the average iterate defined as:

$$\bar{\mathbf{w}} = \frac{1}{T} \sum_{t=1}^T \mathbf{w}_t.$$

How does the convergence behavior of subgradient descent change when we evaluate the performance of $\bar{\mathbf{w}}$ instead of the last iterate?

(a) Show that

$$f(\bar{\mathbf{w}}) - f(\mathbf{w}^*) \leq \frac{1}{T} \sum_{t=1}^T \langle \mathbf{w}_t - \mathbf{w}^*, f(\mathbf{w}_t) \rangle$$

(b) Let $\mathbf{v}_1, \dots, \mathbf{v}_T$ be an arbitrary sequence of vectors. Any algorithm with an initialisation $\mathbf{w}_1 = 0$ and an update rule of the form

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \mathbf{v}_t$$

satisfies

$$\sum_{t=1}^T \langle \mathbf{w}_t - \mathbf{w}^*, \mathbf{v}_t \rangle \leq \frac{\|\mathbf{w}^*\|^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{v}_t\|^2$$

Hint: Prove $\sum_{t=1}^T \langle \mathbf{w}_t - \mathbf{w}^*, \mathbf{v}_t \rangle = \frac{1}{2\eta} (-\|\mathbf{w}_{t+1} - \mathbf{w}^*\|^2 + \|\mathbf{w}_t - \mathbf{w}^*\|^2) + \frac{\eta}{2} \|\mathbf{v}_t\|^2$

(c) Show that in particular if for every $B, \rho > 0$, if for all t we have that $\|\mathbf{v}_t\| \leq \rho$ and if we set $\eta = \sqrt{\frac{B^2}{\rho^2 T}}$ then for every $\mathbf{w}^*$ with $\|\mathbf{w}^*\| \leq B$ we have

$$\frac{1}{T} \sum_{t=1}^T \langle \mathbf{w}_t - \mathbf{w}^*, \mathbf{v}_t \rangle \leq \frac{B\rho}{\sqrt{T}}$$

(d) From the above, prove that if we run the gradient descent (GD) algorithm on f for T steps with $\eta = \sqrt{\frac{B^2}{\rho^2 T}}$, then the output vector $\bar{\mathbf{w}}$ satisfies

$$f(\bar{\mathbf{w}}) - f(\mathbf{w}^*) \leq \frac{B\rho}{\sqrt{T}}.$$

Problem 4

Recall that the (standard) Perceptron algorithm cycles through the data in order (returning to index $t = 1$ after looking at $t = n$) and updates $\theta^{(k)} = \theta^{(k-1)} + y_t x_t$ after the k -th mistake. Recall also that with $\theta^{(0)} = 0$, the number of mistakes is upper bounded by

$$k_{\max} = \frac{R^2 \|\theta^*\|^2}{\gamma^2} \quad (1)$$

under the assumptions $\max_{t=1,\dots,n} \|x_t\| \leq R$ and $\min_{t=1,\dots,n} y_t(\theta^* \cdot x_t) \geq \gamma$. We proved this via the following steps:

- (i) We wrote $(\theta^* \cdot \theta^{(k)})$ in terms of $(\theta^* \cdot \theta^{(k-1)})$ and used recursion to deduce $(\theta^* \cdot \theta^{(k)}) \geq k\gamma$.
- (ii) We wrote $\|\theta^{(k)}\|^2$ in terms of $\|\theta^{(k-1)}\|^2$ and used recursion to deduce $\|\theta^{(k)}\|^2 \leq kR^2$.
- (iii) We used the Cauchy-Schwarz inequality $(\theta^{(k)} \cdot \theta^*) \leq \|\theta^{(k)}\| \cdot \|\theta^*\|$ to show that the previous two findings are only consistent for $k \leq k_{\max}$.

Now suppose that we perform variable step-size updates of the form $\theta^{(k)} = \theta^{(k-1)} + \eta^{(k)} y_t x_t$, where $\eta^{(1)}, \eta^{(2)}, \dots$ is some fixed sequence of positive step sizes.

(a) Follow the above three steps (again under the choice $\theta^{(0)} = 0$) to show that:

- (i) $\gamma \sum_{i=1}^k \eta^{(i)}$,
- (ii) $\|\theta^{(k)}\|^2 \leq \sum_{i=1}^k (\eta^{(i)})^2 R^2$, and
- (iii) $1 \geq \frac{(\sum_{i=1}^k \eta^{(i)})^2}{\|\theta^*\|^2 \sum_{i=1}^k (\eta^{(i)})^2 R^2}$.

(b) For fixed $\gamma > 0$ and $\theta^* \in \mathbb{R}^d$, does the final bound in part (a) guarantee that for R arbitrarily large (but still constant throughout the course of the algorithm), the algorithm eventually stops making mistakes when $\eta^{(i)} = 1$ for all i ? How about when $\eta^{(i)} = \frac{1}{i^2}$? Explain.

(c) Show that we **cannot** use the final bound in part (a) to obtain a better guarantee on the number of mistakes than Equation (1).

(Hint: Although seemingly unrelated, the formula $\text{Var}[Z] = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2$ might be useful.)

(d) Now consider the special case of a constant step size, $\eta^{(i)} = \eta > 0$ for all i . We know from part (c) that we can't expect a better worst-case bound than that of $\eta = 1$. But could $\eta \neq 1$ still lead to faster convergence on a particular data set with $\theta^{(0)} = 0$? Is the answer to this question impacted by the choice of $\theta^{(0)}$?